Dissertation on

"Title of the Capstone Project"

Submitted in partial fulfilment of the requirements for the award of the degree of

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in

Computer Science & Engineering

UE23CS320B – Capstone Project Phase - 2

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Submitted by:

PES1UG23CSXXX Name 1
PES1UG23CSXXX Name 2
PES1UG23CSXXX Name 3
PES1UG23CSXXX Name 4

Under the guidance of

Prof. Guide Name

Designation, Department of CSE

PES University

January - May 2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FACULTY OF ENGINEERING

PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)

100 feet Ring road, BSK 3rd stage, Hosakerehalli, Bengaluru – 560085

PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)

100ft Ring Road, Bengaluru – 560 085, Karnataka, India

FACULTY OF ENGINEERING

CERTIFICATE

This is to certify that the dissertation entitled

'Title of the Capstone Project'

is a bonafide work carried out by

PES1UG23CSXXX Name 1
PES1UG23CSXXX Name 2
PES1UG23CSXXX Name 3
PES1UG23CSXXX Name 4

in partial fulfilment for the completion of sixth-semester Capstone Project Phase - 2 (UE23CS320B) in the Program of Study – **Bachelor of Technology in Computer Science and Engineering** under rules and regulations of PES University, Bengaluru during the period January - May 2026. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. The dissertation has been approved as it satisfies the sixth-semester academic requirements in respect of project work.

Signature	Signature	Signature
Prof. Guide Name	**Dr. Shylaja S S**	**Dr. K S Sridhar**
Designation, Department of CSE	Chairperson	Registrar

External Viva

Name of the Examiners	**Signature with Date**
1. _____	_____
2. _____	_____

DECLARATION

We hereby declare that the Capstone Project Phase - 2 entitled **"Title of the Capstone Project"** has been carried out by us under the guidance of **Prof. Guide Name**, Designation, Department of CSE and submitted in partial fulfilment of the course requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester January - May 2026. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

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PES1UG23CSXXX	**Name 1**	**<Signature>**
PES1UG23CSXXX	**Name 2**	**<Signature>**
PES1UG23CSXXX	**Name 3**	**<Signature>**
PES1UG23CSXXX	**Name 4**	**<Signature>**

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ABSTRACT

[Replace with your abstract. Two paragraphs maximum, per guidelines. Paragraph 1: what problem, why it matters, what this work does. Paragraph 2: the core approach and the headline result.]

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CHAPTER 1

INTRODUCTION

[Replace this text.] Open with the real-world context: what problem exists, who suffers from it, and why current practice is inadequate. Two or three sentences is enough.

[Replace this text.] State what this project does at a high level. One crisp paragraph. Name the core technical approach in plain language (e.g. "a deep-learning pipeline for X") without drowning in jargon.

[Replace this text.] Close with what this Phase 2 report contains and what the reader will find in the remaining chapters. One short paragraph.

CHAPTER 2

LITERATURE SURVEY

2.1 Paper Title Goes Here

Authors: Author et al. **Year:** 20XX **Venue:** Journal / Conference

[Replace.] One paragraph summarising what this paper does, the dataset used, and the headline result. Do not copy from the paper — paraphrase.

[Replace.] One paragraph on limitations of the paper and how your project addresses or extends them.

2.2 Second Paper Title

[Replace with content, structured like the section above.]

2.3 Research Gaps

[Replace.] Summarise the gaps identified across the surveyed work that motivate your project's contribution. Bullet points are acceptable here.

- *[Gap 1]*
- *[Gap 2]*
- *[Gap 3]*

CHAPTER 3

ARCHITECTURE WITH EXPLANATION

3.1 System Overview

[Replace.] In two or three sentences, describe the whole system as a pipeline: input → processing stages → output.

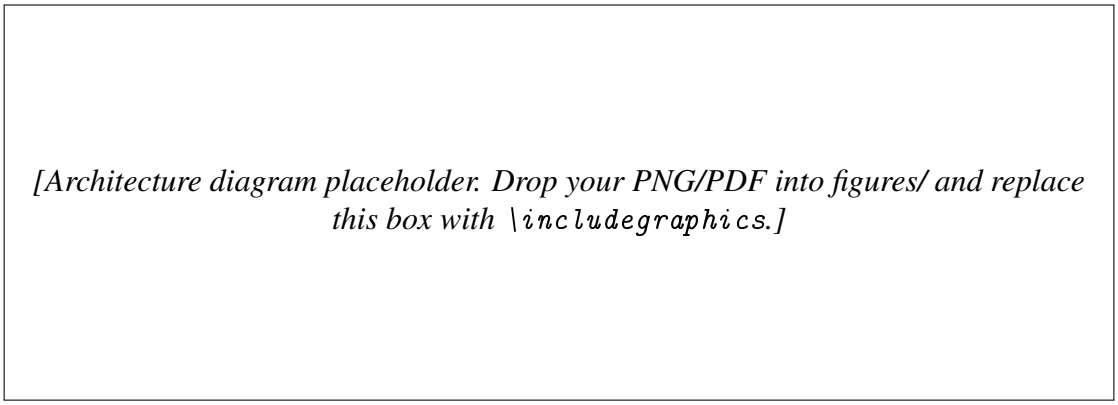


Figure 3.1: End-to-end system architecture.

3.2 Component Description

[Replace.] Walk through each block of Figure 3.1 in order. One short paragraph per block. State what it takes as input, what it produces as output, and one sentence on why it exists.

3.2.1 Component 1

[Replace.]

3.2.2 Component 2

[Replace.]

3.2.3 Component 3

[Replace.]

3.3 Design Rationale

[Replace.] Why this architecture and not an alternative? Name one or two alternatives you considered and state the tradeoff that led you to the chosen design. This is where reviewers look for engineering judgement.

CHAPTER 4

METHODOLOGY

4.1 Dataset

[Replace.] Source, size, licence, class distribution. Explicitly state the train / val / test split strategy and rationale. If you used multiple datasets, describe each and justify the combination.

Table 4.1: Dataset summary.

Split	Images	Positive class %	Notes
Train	–	–	–
Val	–	–	–
Test	–	–	–

4.2 Preprocessing

[Replace.] Resizing, normalisation, contrast enhancement (e.g. CLAHE), denoising. State the exact parameters actually used, not generic defaults.

4.3 Model Architecture

[Replace.] Which model family, which backbone, which hyperparameters. If you used an off-the-shelf architecture, say so and cite it.

4.4 Training Strategy

[Replace.] Optimiser, loss function, learning-rate schedule, batch size, early-stopping criterion, class-weighting if any. Include the exact epoch count and hardware used.

4.5 Evaluation Metrics

[Replace.] Name the metrics and why each was chosen. For medical or imbalanced problems, do not rely on accuracy alone — justify with precision, recall, F1, macro-F1, IoU, Dice, etc. as applicable.

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4.1}$$

CHAPTER 5

IMPLEMENTATION

5.1 Technology Stack

[Replace.] Languages, frameworks (PyTorch/TensorFlow/etc.), libraries, deployment target. Include versions for anything version-sensitive.

5.2 Module 1: *[Name]*

[Replace.] One paragraph on what this module does and how it fits into the architecture shown in Chapter 3.

5.3 Module 2: *[Name]*

[Replace.]

5.4 Module 3: *[Name]*

[Replace.]

5.5 Integration and Pipeline Orchestration

[Replace.] How the modules are chained at runtime. Input format at each boundary, error handling, and any caching or state between stages.

CHAPTER 6

RESULTS

6.1 Quantitative Results

[Replace.] Open with the headline number in one sentence (e.g. "Our best model achieves an F1 of 0.XX on the held-out test set."). Then present the full results table.

Table 6.1: Performance comparison across model variants.

Model	Accuracy	Precision	Recall	F1
Baseline	–	–	–	–
Variant A	–	–	–	–
Variant B	–	–	–	–
Ours	–	–	–	–

6.2 Qualitative Results

[Replace.] Show representative outputs. Include at least one failure case — this is the single easiest way to demonstrate that you understand your own system's limits.

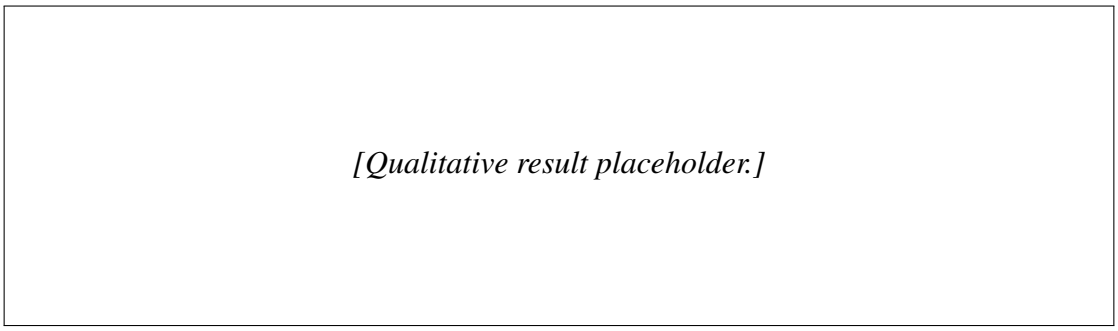


Figure 6.1: Representative output from the proposed system.

6.3 Ablation / Comparative Analysis

[Replace. Optional but strongly recommended.] Which design choices mattered? Remove one component at a time and report the delta in the headline metric.

6.4 Discussion

[Replace.] Brief. Two paragraphs at most. What do the numbers mean, and what are the most honest limitations of the evaluation itself (e.g. single-centre data, small test set)?

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Summary of Phase 2

[Replace.] A paragraph on what this phase delivered. Tie each deliverable to the chapter where it is described (Architecture Chapter 3, Implementation Chapter 5, Results Chapter 6).

7.2 Key Contributions

- *[Contribution 1 — one line.]*
- *[Contribution 2 — one line.]*
- *[Contribution 3 — one line.]*

7.3 Limitations

[Replace.] Be honest. If a limitation exists and you hide it, examiners will find it anyway and will trust the rest of the report less.

7.4 Future Work

[Replace.] Concrete next steps, not wishlist items. Each bullet should be something a successor could pick up and act on.

- *[Future item 1.]*
- *[Future item 2.]*
- *[Future item 3.]*

REFERENCES

BIBLIOGRAPHY

[1] W. M. Dorsey, A. Stumme, K. M. Charipar, and N. A. Charipar, “3-D-Printed Circular Array for WiMAX Base Station,” *IEEE Antennas and Wireless Propagation Letters*, vol. 18, no. 6, pp. 1159–1163, Jun. 2019, doi: 10.1109/LAWP.2019.2911354.